



Some Open Source Anti-Covid Solutions

The second wave of Covid-19 outbreak has created a critical situation for us all. Thankfully, the open source community and several groups of enthusiastic techies have taken up the challenge and are offering some excellent solutions for combating the coronavirus.

The open source community is also providing a low-cost design for polymerase chain reaction (PCR) testing. The entire device can be made for about 50 dollars (approx. 3,700 rupees) only.

EFY team has reviewed and shortlisted some of the open source products. Hope you will find the information useful in these difficult times.

Pocket PCR

As we know, the PCR test is one of the reliable solutions for covid detection. This Pocket PCR is an open source testing device that is small and affordable but powerful like a full-fledged thermocycler with large laboratory machine-like features. Even in this small size, the device features PID temperature control with great accuracy. It has 5 × 0.2ml PCR tubes while its overall size is 33mm × 105mm × 50mm and weighs just 50 grams. It can be powered with any 2A power adaptor or computer port.

For details: https://github.com/GaudiLabs/PocketPCR/tree/master/PCB_Layouts/PocketPCR_PCB/RocketPCR
<http://gaudi.ch/PocketPCR>



OpenPCR

OpenPCR is a small and affordable PCR machine that features an adjustable-height heated lid. This enables heating of the top of the tube up to 120 degrees Celsius, preventing any substantial condensation. It includes an intelligent GUI based control system that helps the device to run using any C (connected via USB). The GUI helps to program the PCR protocol. OpenPCR allows flexibility of disconnecting after being programmed. The machine will, however, continue to carry out the reaction while an LCD shows its live status with estimated remaining time.

For Arduino code and details: <https://openpcr.org>
<https://github.com/jperfetto/OpenPCR>

UV Robot

This UV robot sanitises large areas such as hospitals, offices, halls, and hotel rooms without creating any risk of exposure to dangerous UVC light. Unlike other UVC robots, this device gives wireless access to its GUI with a real-time video feed (for controlling the UV robot).

For details: <https://www.electronicsforu.com/electronics-projects/hospital-sanitizing-robot>

